

Assignment-Regression Algorithm

Problem Statement or Requirement:

A client's requirement is, he wants to predict the insurance charges based on the several parameters. The Client has provided the dataset of the same.

As a data scientist, you must develop a model which will predict the insurance charges.

- 1.) Identify your problem statement
 - Machine Learning
 - Supervised Learning
 - Regression
- 2.) Tell basic info about the dataset
 - Total number of rows: **1338**
 - Total number of columns: **6**
- 3.) Mention the pre-processing method if you're doing any
 - Yes, we are doing the preprocessing data converting string into numerical data → ordinal data
- 4.) Develop a good model with r^2_score . You can use any machine learning algorithm; you can create many models. Finally, you have to come up with final model.
 - Created model using random forest algorithm.
- 5.) All the research values (r^2_score of the models) should be documented. (You can make tabulation or screenshot of the results.)

Algorithm to find the R^2_score Using Hyper Tuning Parameter

1. **Multiple Linear Regression** → $R^2 = 0.7895$
2. **Support Vector Machine**

S.NO	HYPER PARAMETER	r VALUE			
		LINEAR	POLY	RBF	SIGMOID
1	C0.01	-0.0765	-0.2570	-0.0897	-0.0895
2	C10	113.0486	-1163348.2386	-0.0804	0.0131
3	C100	-146.1431	-32979013.9815	-0.0213	-0.5436
4	C500	-152.2224	-29631241.4196	-6.2393	-5.4844
5	C1000	-154.8728	-10543590.2291	-0.0159	-4.5499

SVM use sigmoid and $C=10 \rightarrow R^2=0.0131$

3. Decision Tree

S.NO	CRITERION	SPLITTER	MAX FEATURES	r_VALUE
1	squared_error	best	None	0.6957
2	squared_error	best	log2	0.6755
3	squared_error	best	sqrt	0.6755
4	squared_error	random	None	0.7210
5	squared_error	random	log2	0.6969
6	squared_error	random	sqrt	0.6969
7	absolute_error	best	None	0.6476
8	absolute_error	best	log2	0.7225
9	absolute_error	best	sqrt	0.7225
10	absolute_error	random	None	0.7180
11	absolute_error	random	log2	0.7116
12	absolute_error	random	sqrt	0.7116
13	poisson	best	None	0.7288
14	poisson	best	log2	0.7553
15	poisson	best	sqrt	0.7553
16	poisson	random	None	0.7019
17	poisson	random	log2	0.6261
18	poisson	random	sqrt	0.6261

Decision Tree use poisson, best, log2 or sqrt $\rightarrow R^2=0.7553$

4. Random Forest

S.NO	N_ESTIMATORS	CRITERION	MAX FEATURES	r_VALUE
1	100	squared_error	None	0.8540
2	100	squared_error	log2	0.8711
3	100	squared_error	sqrt	0.8711
4	100	absolute_error	None	0.8498
5	100	absolute_error	log2	0.8692
6	100	absolute_error	sqrt	0.8692
7	100	poisson	None	0.8526
8	100	poisson	log2	0.8679
9	100	poisson	sqrt	0.8679

Random forest use n_estimators=100, squared_error, log2 or sqrt $\rightarrow R^2=0.8711$

6.) Mention your final model, justify why u have chosen the same

- Final model $\rightarrow$ random forest algorithm which gives R^2 value near to 1